POOJA D ANCHAN

AI & LLM ENGINEERING | CLOUD & DEVOPS | STORAGE PLATFORM QA & AUTOMATION

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PROFESSIONAL SUMMARY

Software engineer with 8 years of experience in Storage Platform Validation & QA, high-availability systems, release validation, and test automation, complemented by 3 years of hands-on Cloud and DevOps project implementation. Now focused on AI and LLM Engineering, with practical work across RAG, AI agents, fine-tuning, embeddings, vector search, tokenization, structured outputs, and AI evaluation. Brings strong systems thinking, debugging, automation, cloud-native engineering, and a quality-first mindset to building reliable AI solutions.

CORE TECHNICAL SKILLS

AI / LLM Engineering: Python, PyTorch, Hugging Face Transformers, fine-tuning, NLP, prompt engineering, structured outputs, RAG, multimodal RAG, embeddings, semantic search, vector databases, AI agents, Google ADK, Gemini, evaluation, tokenization

Data / APIs: PostgreSQL, pgvector, Pinecone, FastAPI, REST APIs, Pydantic, LangChain, Sentence-Transformers

Cloud / DevOps: AWS (EKS, EC2, S3, Lambda, SNS, IAM), Azure, Docker, Kubernetes, Terraform, ArgoCD, GitOps, Jenkins, GitHub Actions, Ansible, Linux, Bash

QA / Automation: Platform Validation, High Availability, Reliability Testing, Test Automation, Root-Cause Analysis, Perl, pytest, Ruff, mypy, Bandit, SonarQube, ELK, Nagios

SELECTED AI & LLM ENGINEERING PROJECTS

Secure EHR Insight & Clinical Validator

Python | PostgreSQL / pgvector | BioClinical ModernBERT | Microsoft Presidio | FastAPI | Streamlit | Docker | AWS EC2 | Gemini

- Built a privacy-aware clinical semantic-search and RAG pipeline over MIMIC-IV transcripts; generated 768-dimensional biomedical embeddings locally and stored 10,000 records in PostgreSQL/pgvector for cosine-similarity retrieval.
- Implemented PII redaction with Microsoft Presidio and designed API/UI and guardrail layers for grounded clinical responses while keeping sensitive vector data within PostgreSQL.

Kubernetes Investigation Agent

Google ADK | Gemini 2.5 Flash | Python | Kubernetes | pytest

- Built an AI-assisted Kubernetes incident investigation agent that parses CRI-O, JSON, Kubelet, and multiline logs and identifies OOMKilled, CrashLoopBackOff, DNS, RBAC, storage, configuration, and dependency failures.
- Combined deterministic root-cause analysis with Gemini via Google ADK to reconstruct incident timelines, cite supporting evidence, and generate evidence-grounded remediation guidance; added unit and integration tests.

Transformer Fine-Tuning Portfolio

Hugging Face | PyTorch | Transformers | Datasets | CUDA | Google Colab

- Fine-tuned DistilBERT for sentiment classification, DistilGPT2 for AG News classification, and FLAN-T5 Large for abstractive summarization; published trained models to Hugging Face Hub.
- Practiced architecture-specific tokenization/evaluation, full fine-tuning, gradient accumulation, checkpointing, and GPU-aware training; achieved 89.8% accuracy on DistilGPT2 news classification and ROUGE-1 ~0.42 on FLAN-T5 summarization.

Advanced & Multimodal RAG

Python | Pinecone | Embeddings | Vector Search | Document & Image Retrieval | Jupyter

- Built a multimodal RAG workflow over a company report by extracting document text and images, generating embeddings, indexing content in Pinecone, and retrieving grounded context for question answering.
- Explored chunking, retrieval depth, overlap, reranking/query-rewriting concepts, and retrieval/generation evaluation for improving RAG quality.

Additional AI Engineering Portfolio

- Medical BPE tokenizer training (16K vocabulary, PubMed corpus), multilingual/token-fertility experiments, structured LLM outputs with Pydantic/Instructor/Outlines, prompt-engineering notebooks, and RAG/customer-support prototypes. Portfolio links: GitHub and Hugging Face above.

CAREER BREAK & PROFESSIONAL DEVELOPMENT

Dubai, UAE | Jul 2018 - Present

- Career break for parenting and relocation; since 2023, actively preparing to return to technology through structured learning, certifications, and hands-on engineering projects.
- Built and deployed Cloud/DevOps projects across AWS, Azure, Kubernetes, Docker, Terraform, Jenkins, ArgoCD, GitOps, Ansible, Python, and CI/CD workflows.
- Since 2026, focused deeply on AI and LLM Engineering through live, hands-on Advanced Route-Production AI & LLM Engineering classes and portfolio projects covering RAG, agents, fine-tuning, tokenization, embeddings, evaluation, and AI application architecture.

PROFESSIONAL EXPERIENCE

Module Lead - Platform QA | Mindteck (NetApp Storage Platform Validation)

Bengaluru, India | May 2015 - Jun 2018

- Led major test-automation initiatives for NetApp storage platforms, owning test strategy, planning, lab/infrastructure requirements, execution, and release delivery.
- Designed and maintained Perl/Linux automation frameworks and scripts for platform validation and regression testing across hardware and software releases.
- Performed high-availability, reliability, system, and release validation; partnered with development teams on defect analysis, debugging, root-cause investigation, and release readiness.
- Mentored new team members and delivered technical/domain training on storage platform QA and automation.

Senior Software Engineer - Platform QA | Mindteck (NetApp Storage Platform Validation)

Bengaluru, India | May 2012 - Apr 2015

- Validated CPU, memory, motherboard, power supply, service processor, PAM, NVRAM, BIOS, Loader, SP, BMC, and FPGA firmware releases across platform configurations.
- Executed integration, system, performance, capacity, regression, reliability, and release-candidate validation across release milestones.

Software Engineer - Platform Validation | Mindteck (NetApp Storage Platform Validation)

Bengaluru, India | May 2010 - Apr 2012

- Performed early platform software and firmware validation ahead of formal QA cycles and validated Service Processor and BIOS releases across hardware configurations.
- Developed an end-to-end automation workflow to detect new software builds and automatically update NetApp hardware for test execution.

SELECTED CLOUD & DEVOPS PROJECTS

Python DevOps Automation Toolkit

Python | AWS | pytest | Ruff | mypy | Bandit | GitHub Actions

- Built a production-style automation toolkit for HTTP health checks, environment diagnostics, AWS resource auditing, structured logging, automated testing, security scanning, and CI/CD.

AWS EKS / GitOps Portfolio

AWS EKS | Kubernetes | Terraform | ArgoCD | Docker | GitOps

- Deployed three-tier and microservices applications on AWS EKS, provisioned infrastructure with Terraform, and implemented GitOps-based deployment and synchronization using ArgoCD.

Serverless S3 Event Notification

AWS Lambda | S3 | SNS

- Built an event-driven workflow that triggers Lambda on S3 uploads and sends automated notifications through SNS.

CERTIFICATIONS & PROFESSIONAL TRAINING

- Advanced Route-Production AI & LLM Engineering - live, hands-on program focused on building and architecting LLM systems (Jul 2026 - Present)
- Intellipaat DevOps Certification (Mar 2024)

Microsoft Certifications: Azure Solutions Architect Expert | Azure Administrator Associate | Azure Fundamentals (2023)
Education: B.E., Electronics & Communication Engineering | 2009